1. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

> *"Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation"*

- Highly relevant: This is the core structure of the solution.
- Students must carry out the following clear sequence:
  1. Identify the radius of rotation: Blade length = 40 m → the tip traces a circle of radius $r = 40$ m;
  2. Calculate the circumference: One full revolution covers $C = 2\pi r = 2\pi \times 40$ meters;
  3. Compute distance traveled per minute: At 20 rpm, distance = $20 \times C = 20 \times 2\pi \times 40$ meters/minute;
  4. Convert to distance per hour: Multiply by 60 → $v = 20 \times 2\pi \times 40 \times 60$ meters/hour;
  5. Convert to km/h: Divide by 1000 → $v = \frac{20 \times 2\pi \times 40 \times 60}{1000}$ km/h;
  6. Numerical evaluation: Approximately $\frac{96{,}000\pi}{1000} \approx 301.6$ km/h.
- This complete chain—conceptual modeling → formula application → unit conversion → numerical computation—is a textbook example of Competency #12.

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2. Competency #7: Manipulate numbers, algebraic expressions, equations, and geometric representations

> *"Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations"*

- Highly relevant: The solution involves multiple types of mathematical manipulation:
  - Geometric modeling: Representing the blade tip' s motion as circular (radius = blade length);
  - Algebraic formulation: Constructing the speed expression $v = \text{rpm} \times \text{circumference} \times \text{time conversion}$;
  - Numerical computation: Handling expressions with π and large numbers (e.g., $20 \times 80\pi \times 60$);
  - Unit conversion: Meters → kilometers, minutes → hours (×60 ÷1000).
- The integration of geometry, algebra, arithmetic, and unit handling is essential to solving the problem.

3. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process

> *"Use an understanding of the context to guide or expedite the mathematical solving process..."*

- Strongly relevant: Knowledge of the physical context directly informs the modeling:
  - Recognizing that "blade tip speed" refers to linear (tangential) speed, not angular speed;
  - Knowing that a point on a rotating object follows a circular path with radius equal to the blade length;
  - Understanding that "maximum rotational speed" corresponds to "maximum linear speed," so no variable-speed analysis is needed;
  - Using the standard simplification: one revolution = one circumference traveled (ignoring aerodynamic complexities).
- This context-informed abstraction ensures the model is both physically meaningful and mathematically tractable.

4. Competency #3: Implement a given strategy to determine a mathematical solution

> *"Implement a given strategy to determine a mathematical solution"*

- Moderately relevant: Although no explicit formula is provided, the standard physics approach (linear speed = revolutions per minute × circumference) is the implied strategy;
- Students are expected to recall and faithfully apply this conventional method within a familiar domain.

5. Competency #8: Articulate a solution, showing and/or summarising intermediate mathematical results

> *"Articulate a solution, showing and/or summarising and presenting intermediate mathematical results"*

- Moderately relevant: The prompt explicitly asks students to "describe your solution process," requiring them to present:
  - Radius = 40 m;
  - Circumference = $2\pi \times 40 \approx 251.3$ m;
  - Distance per minute = $20 \times 251.3 = 5{,}026.5$ m/min;
  - Distance per hour = $5{,}026.5 \times 60 = 301{,}592.9$ m/h ≈ 301.6 km/h.
- Showing key intermediate steps is necessary to fulfill the task requirements.

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6. Competency #1: Perform a simple calculation

> *"Perform a simple calculation"*

- Weakly but present: Involves basic operations (multiplication, division, approximation of π), though embedded within a complex workflow rather than standing alone.

1. 第12项：使用多步骤程序得出数学解、结论或一般化结果

> *"Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation"*

- 高度相关：这是解题的核心结构。
- 学生必须执行以下清晰的多步程序：
  1. 识别旋转半径：叶片长 40 m → 尖端轨迹为半径 $r = 40$ m 的圆；
  2. 计算周长：一圈路径长 $C = 2\pi r = 2\pi \times 40$ 米；
  3. 计算每分钟行进距离：最大转速 20 转/分钟 → 每分钟移动 $20 \times C = 20 \times 2\pi \times 40$ 米；
  4. 转换为每小时距离：乘以 60 → $v = 20 \times 2\pi \times 40 \times 60$ 米/小时；
  5. 转换为千米/小时：除以 1000 → $v = \frac{20 \times 2\pi \times 40 \times 60}{1000}$ km/h；
  6. 数值计算：约 $\frac{96000\pi}{1000} \approx 301.6$ km/h。
- 这一概念建模 → 公式应用 → 单位换算 → 数值求解的完整链条，是第12项的典范体现。

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2. 第7项：操作数字、代数表达式、方程及几何表示

> *"Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations"*

- 高度相关：解题涉及多重数学操作：
  - 几何建模：将叶片尖端运动抽象为圆周运动（半径 = 叶片长）；
  - 代数表达：构建速度公式 $v = 转速 \times 周长 \times 时间换算$；
  - 数值计算：处理含 π 的表达式，进行大数乘除（如 $20 \times 80\pi \times 60$）；
  - 单位换算：米 → 千米，分钟 → 小时（×60 ÷1000）。
- 几何、代数与算术的融合操作是本题的关键执行环节。

3. 第13项：利用对情境的理解指导或加速数学求解过程

> *"Use an understanding of the context to guide or expedite the mathematical solving process..."*

- 强相关：物理情境知识直接驱动建模选择：
  - 理解“叶片尖端速度”即线速度，而非角速度；
  - 知道旋转物体上一点的路径是圆，半径等于叶片长度；
  - 明白“最大转速”对应“最大线速度”，无需考虑变速；
  - 采用每转一圈走一个周长的简化模型（忽略空气动力学等复杂因素）。
- 这种基于现实机制的合理抽象使数学模型既准确又可解。

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4. 第3项：实施给定策略以确定数学解

> *"Implement a given strategy to determine a mathematical solution"*

- 中等相关：虽然题目未明确给出公式，但标准物理方法（线速度 = 转速 × 周长）是隐含策略；
- 学生需调用这一通用策略并忠实执行计算流程，属于在熟悉领域内实施常规方法。

5. 第8项：清晰表述解法，展示中间数学结果

> *"Articulate a solution, showing and/or summarising and presenting intermediate mathematical results"*

- 中等相关：题目明确要求“描述解题过程”，意味着必须呈现：
  - 半径 = 40 m；
  - 周长 = $2\pi \times 40 \approx 251.3$ m；
  - 每分钟距离 = $20 \times 251.3 = 5026.5$ m/min；
  - 每小时距离 = $5026.5 \times 60 = 301,592.9$ m/h ≈ 301.6 km/h。
- 展示关键中间步骤是满足题目要求的必要条件。

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6. 第1项：进行简单计算

> *"Perform a simple calculation"*

- 弱相关但存在：包含基础运算（乘法、除法、π近似），但嵌套在复杂流程中，非独立技能。